

CS & IT ENGINEERING

Algorithms

Greedy Method

Lecture No.- 04



By- Aditya sir

Recap of Previous Lecture



Topic

Topic

JSD
OMP Intro

Topics to be Covered



Topic

Topic

OMP

Topic

Huffman Encoding



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on LinkedIn where I share my insights and guide students and professionals.



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TTTe



Topic : Greedy Algorithms

P3: Optimal merge patterns:- (Application of merging algo.)

Given 'n' files

It is ~~need~~^{need} to merge them using 2-way merging, such that the total number of record moments are minimum.

Default:- [Merging 2-files at a time]



Topic : Greedy Algorithms

Concept:-

Given 2 sorted files 'A' & 'B' having 'n' and 'm' records respectively. To merge them into a single sorted file, it requires $(n + m)$ record movements.



Topic : Greedy Algorithms

Eg.2.

$n = 3$

$m = 4$

A: [2, 3, 5]

B: [1, 4, 8, 10]

[1, 2, 3, 4, 5, 8, 10,]

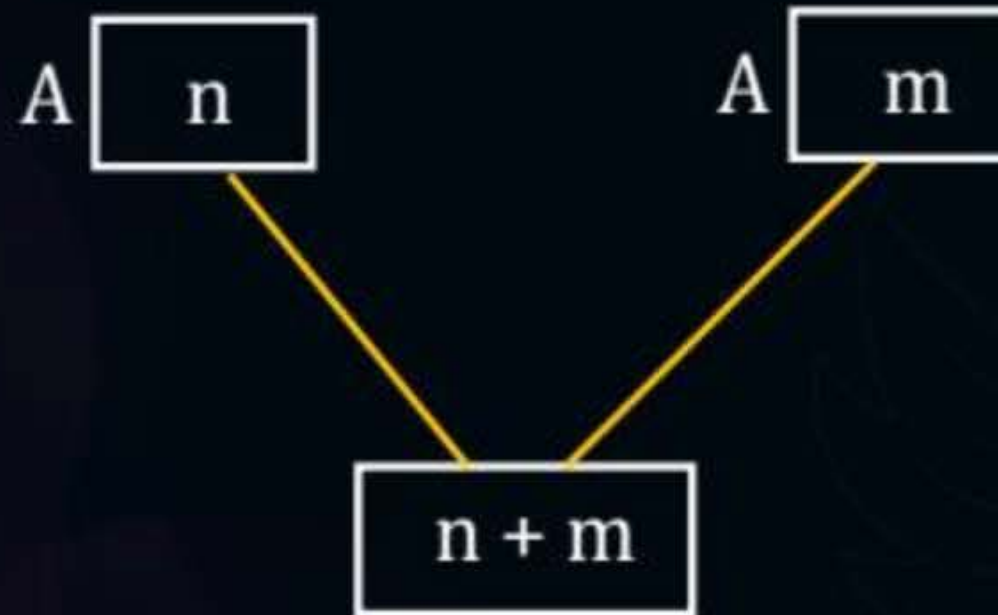
$$\begin{aligned}\text{Record movement} &= n + m \\ &= 3 + 4 \\ &= 7\end{aligned}$$



Topic : Greedy Algorithms

2-way merging Algorithm (merge sort)

Number of elements comparisons:



Best case = $\min(n, m)$

Worst case = $(m + n - 1)$ element comparisons.



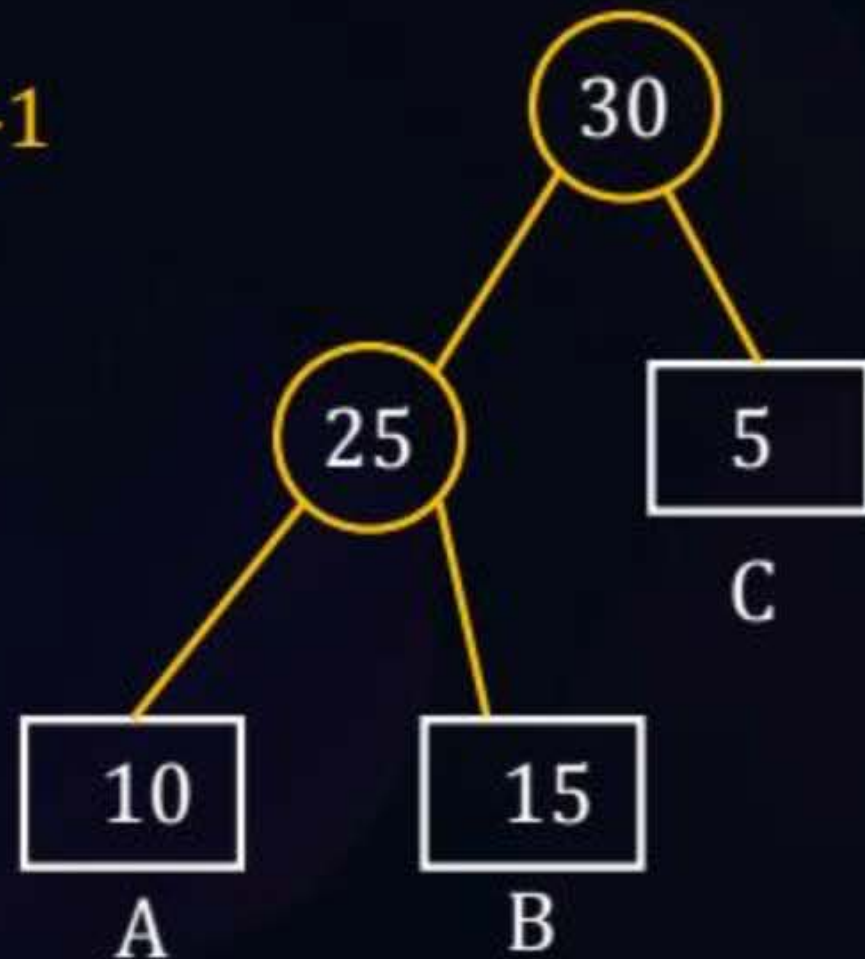
Topic : Greedy Algorithms

Eg.3. A: 10 $\rightarrow$ “number of records in files A”

B: 15 $\rightarrow$ “number of records in files B”

C: 5 $\rightarrow$ “number of records in files C”

Case-1



Total no. of records movement = $25 + 30$
 $= 55$

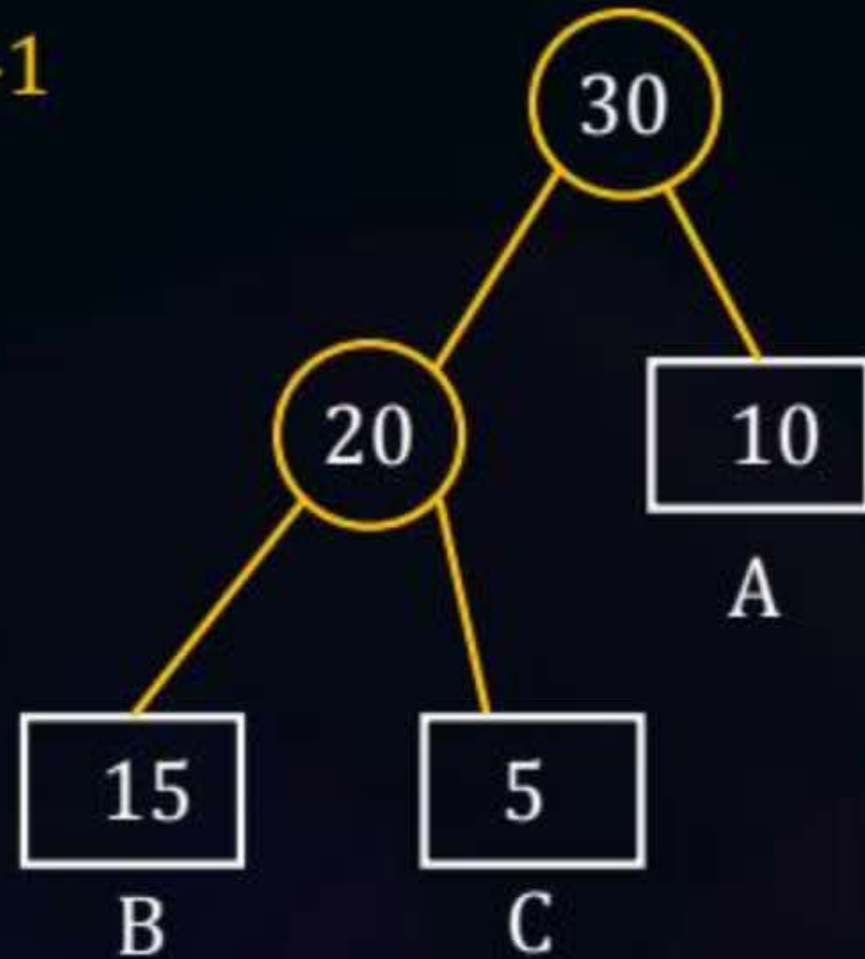
Order = ABC



Topic : Greedy Algorithms

α - Way binary merge tree

Case-1



Total no. of records movement = $20 + 30$
 $= 50$

Order = BCA



Topic : Greedy Algorithms

Observation.

1. Ordering matters and ^{impacts} ~~inputs~~ the total number of records movements.
2. For 'n' files, how many times 2-way merging operation is required ? $\Rightarrow (n-1)$
3. Objective:- Minimize the total number of records movements.



Topic : Greedy Algorithms

Observation.

4. Size of solution space = ?

$n!$ arrangement

All $n!$ Solution are feasible.

ABC
BCA
ACB
BAC
CAB
CBA

↓

$$\underline{A} \underline{B} \underline{C} \quad 3! = 6$$



Topic : Greedy Algorithms

Appr.-1

Brute force/ Enumeration

- ^{guarantees} ~~Quarants~~ optimal sol.
- But time complexity = $O(n!)$
 $= O(n^n)$

Hence, we need a based solution → greedy algo.



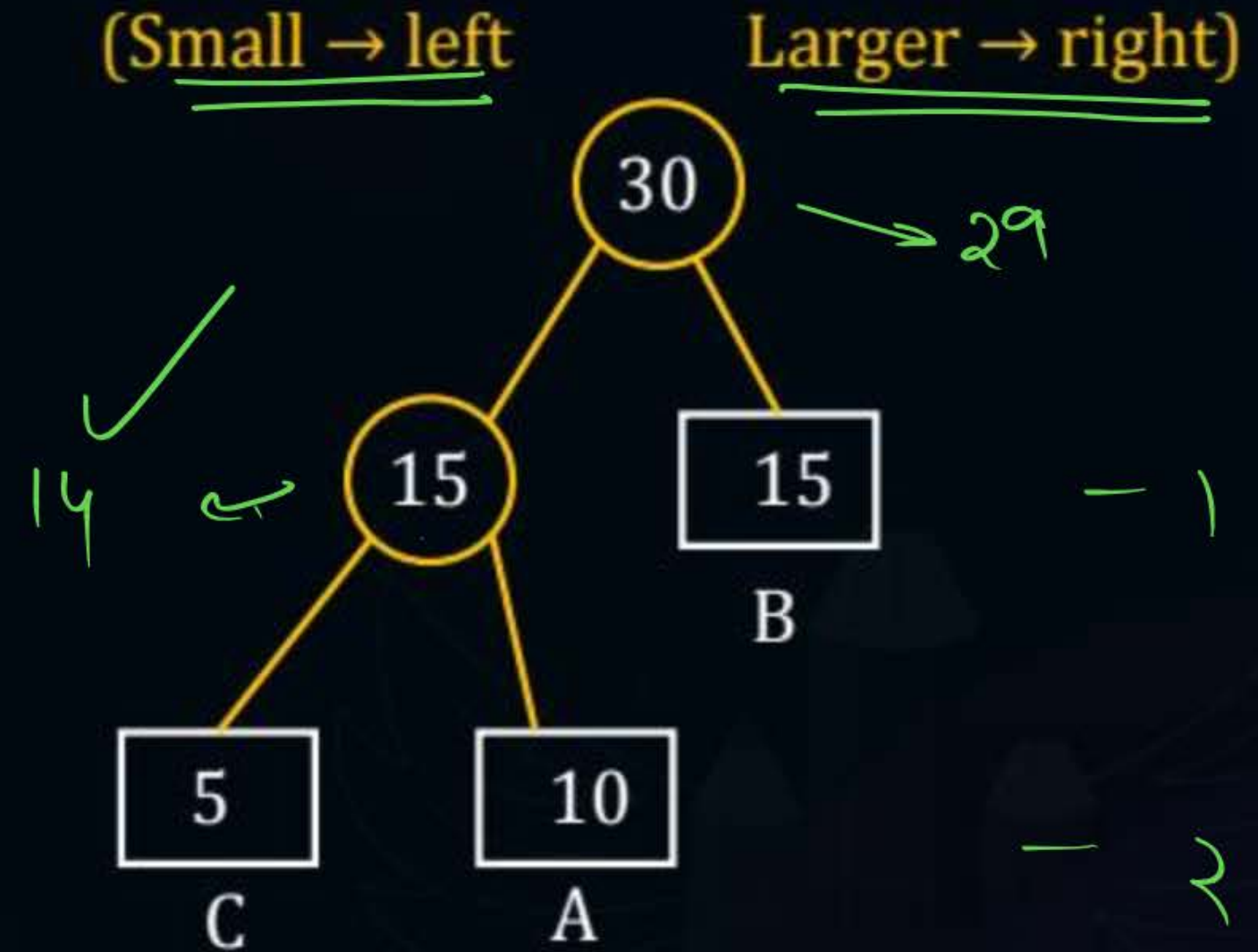
Topic : Greedy Algorithms

Appr.-2

Greedy algorithm

E.g. A : 10 — ~~2~~
B : 15 — 3 15
C : 5 — ~~1~~

Total number of record movement
= 15 + 30
= 45 → optimal



Total elements -verse comparison
= (15 - 1) + (30 - 1)
= 14 + 29 = 43
= 43



Topic : Greedy Algorithms

#Q. $F1 \rightarrow 12$

$F2 \rightarrow 5$

$F3 \rightarrow 18$

$F4 \rightarrow 10$

$F5 \rightarrow 2$

$RM = ?$



Topic : Greedy Algorithms

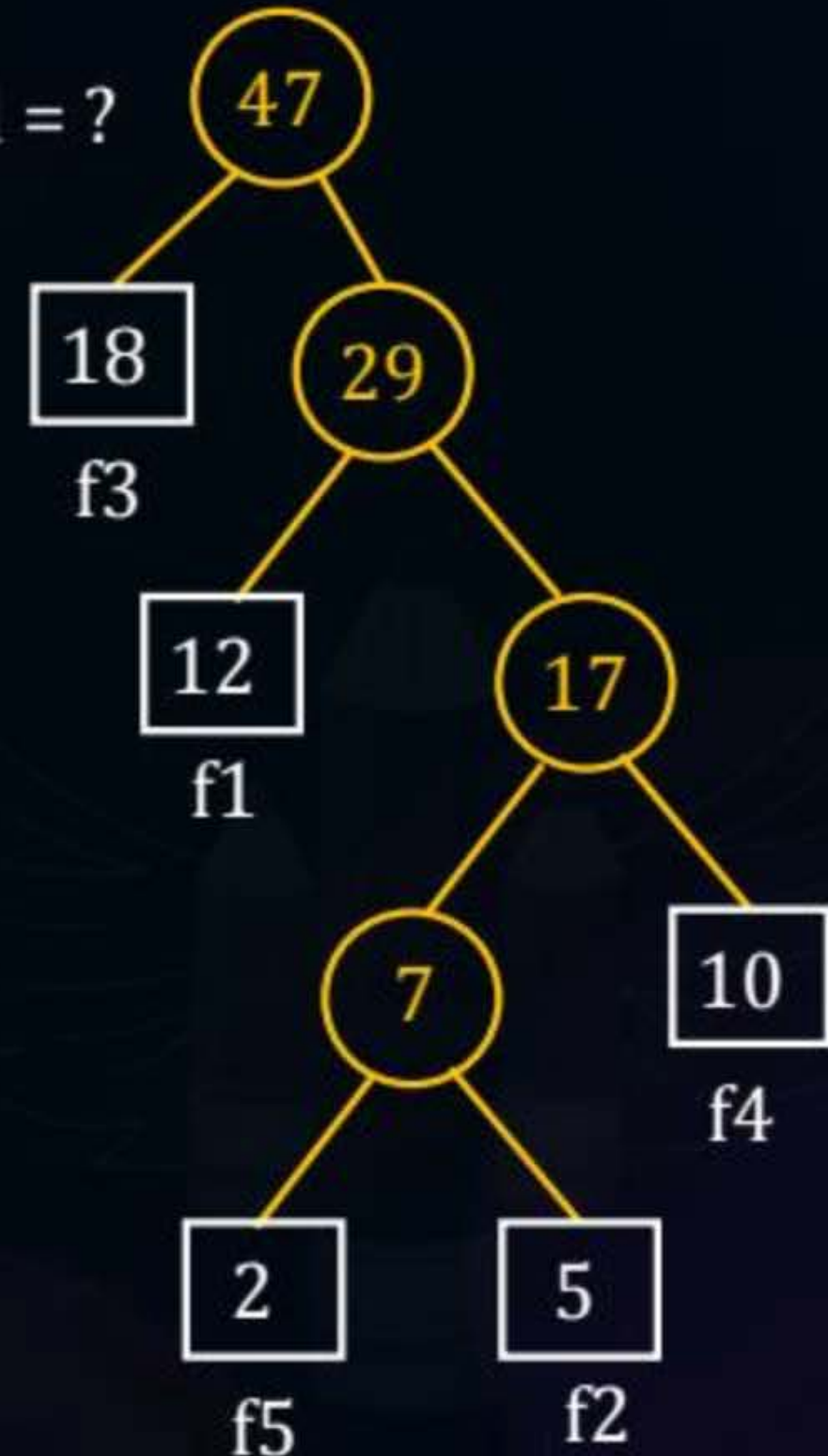
Minimum number of records movement required = ?

$$\begin{aligned}\text{Total records movement} &= 7 + 17 + 29 + 47 \\ &= 24 + 29 + 47 \\ &= 53 + 47 \\ &= 100\end{aligned}$$

Handwritten notes showing the selection of records for merging:

- $F_1 = 4$ (crossed out)
- $F_2 = 2$ (circled)
- $F_3 = 18$ (crossed out)
- $F_4 = 10$ (crossed out)
- $F_5 = 1$ (circled)

Other numbers shown: 5, 17, 29, 7, 12, 18.





Topic : Greedy Algorithms

Formally

The total number of record movement = weighted external path length of the 2 – way binary merge tree.

$$= \sum_{i=1}^n d_i * q_i$$

Where

d_i = distance from root to file i

q_i = size/freq. of file





Topic : Greedy Algorithms

Greedy algo for optimal merge patterns

```
tree node = record  
{  
    tree node * lchild;  
    tree nodes * rchild ;  
    inf weight;  
}
```




Topic : Greedy Algorithms

Algo tree (n)

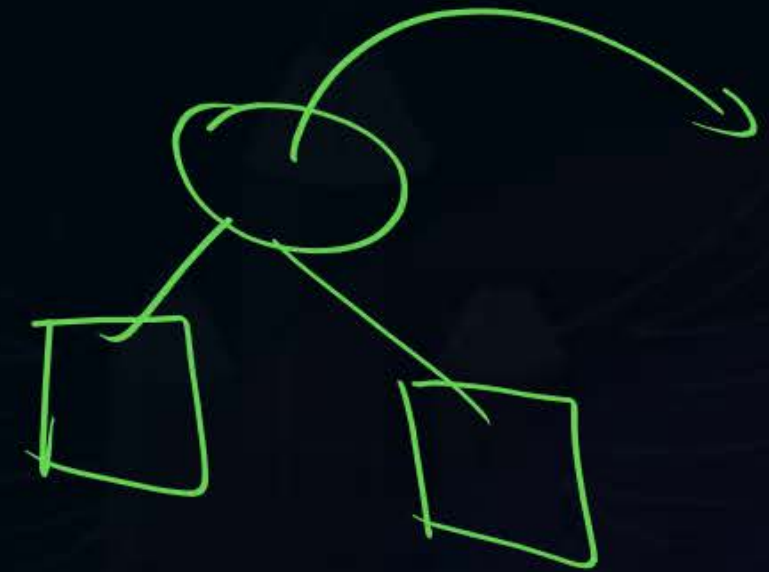
{

// list is a global list of n single nodes binary tree, as described before
for i : 1 to (n-1) ; $\rightarrow O(n)$

{

1. Pt = new (treenode)
2. pt $\rightarrow$ lchild = least (list)
3. pt $\rightarrow$ rchild = least (list)
4. pt $\rightarrow$ weight = (l pt $\rightarrow$ lchild) $\rightarrow$ weight
+ ((pt $\rightarrow$ rchild) $\rightarrow$ weight)
5. insert (list, pt)

}





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Time complexity: (for 'n' files)

1. Linear linked list:

('n' elements)

Insert $\rightarrow O(1)$

least $\rightarrow O(n)$

Overall TC; $O(n^2)$



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Implementation of list

2. Heap (non-linear)

('n' elements)

Insert

Delete

Dec./Inc.

} $\rightarrow O(\log n)$

Min Heap: least $\rightarrow O(1)$

Overall TC: $O(n * \log n)$

4) Huffman Encoding



Topic : Greedy Algorithms

3. Huffman Encoding

(Application of optimal merge pattern)

(Data encoding and data compression Technique)

(representing some elements with a unique code)

Compression → (reduce size of data without data loss)



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Eg.1. 4 Character /elements

w, x, y, z

Straight forward Binary encoding

Normal Encoding $W \rightarrow 0\ 0$

Uniform Encoding $X \rightarrow 0\ 1$

$Y \rightarrow 1\ 0$

$Z \rightarrow 1\ 1$



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If we use uniform encoding:

1. 6 elements $\rightarrow$ 3 bits
2. 10 elements $\rightarrow$ 4 bit $\rightarrow \log_2 10 \rightarrow \lceil 3. x \rceil$
 $\rightarrow 4$

4 bits $\rightarrow$ max $2^4 = 16$ elements can be uniquely encoded

3. 17 element $\rightarrow$ 5 bits.....
4. 100 elements $\rightarrow$ 7 bits/ elements or character.

16

10 ✓



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In general

To represent N elements using uniform encoding
we need $\rightarrow \underline{\underline{\lceil \log_2 (N) \rceil \text{ bit/character.}}}$

Imp. Observations:

1. Using uniform encoding data compression is not possible.
2. Hence, for data compression we need non-uniform encoding.



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Huffman Encoding (non-uniform encoding)

V. Imp

Idea:- Encoding/code → assigned based on the frequency of that character elements.

Logic:-

$$\left[\text{(no. of bits to encode)} \times \frac{1}{\text{(freq. of character)}} \right]$$

1. Assign more bits to low frequently occurring elements.
2. Assign less bits to high frequently occurring elements.
3. No. 2 elements/characters will have tree same encoding (as if needs to be unique).

aabcaac



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Prob

A $\rightarrow$ 0.23.....(4)

B $\rightarrow$ 0.05.....(1)

C $\rightarrow$ 0.48.....(5)

D $\rightarrow$ 0.15.....(3)

E $\rightarrow$ 0.09.....(2)

Priority (low to high)





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Eg..

Given: Language/ Elements $\Rightarrow \{A,B,C,D,E\} \rightarrow 5$ character

If uniform encoding: $\lceil \log_2 N \rceil$ bits/character

$$= \lceil \log_2 5 \rceil$$

$$= 3 \text{ bit / character}$$

~~Note:- Start from lowest level.~~



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2-way Optimal Binary merge tree:

Left Branch $\rightarrow 0$

Right Branch $\rightarrow 1$

Standard

Left $\rightarrow$ Smaller

Right $\rightarrow$ Larger

$$\begin{array}{r} 1 \\ + 0.52 \\ + 0.29 \\ + 0.14 \\ \hline 1.95 \end{array}$$





Topic : Greedy Algorithms

A $\rightarrow$ 10 $\rightarrow$ 2 bits

B $\rightarrow$ 1100 $\rightarrow$ 4 bits

C $\rightarrow$ 0 $\rightarrow$ 1 bits

D $\rightarrow$ 111 $\rightarrow$ 3 bits

E $\rightarrow$ 1101 $\rightarrow$ 4 bits

Non-Uniform encoding & all are
unique



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#Q. ~~Average~~ number of bit required using Huffman encoding? Coding:

Avg

= Weighted external path length:

$$= \underline{\underline{\sum_{i=1}^n d_i * q_i}}$$

$$= \underline{\underline{1.95 \text{ bits/char}}}$$



Topic : Greedy Algorithms

Type 3:-

#Q. "CCACCCA BDC"

How many bit required to above text using:-

1. Uniform Encoding:

$$\rightarrow 10 \times 3 = \underline{30 \text{ bits}}$$

2. Huffman Encoding:

(Non-uniform)

HW



2 mins Summary



Topic

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Topic

Topic

Topic

OMP
Huffman Encoding



THANK - YOU